



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Metody uczenia reprezentacji dla generalizacji modeli uczenia głębokiego między symulacją a rzeczywistością w fizycznej sztucznej inteligencji / Representation learning methods for simulation-to-reality generalization of deep learning models in physical AI

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	Doctoral School courses. Reading the literature and considering possible research directions
Semestr 2	Doctoral School courses and literature review. Choosing the dissertation topic with the supervisor and preparing the Individual Research Plan
Semestr 3	Preparing the research environment with digital twins, simulators and public datasets. Initial experiments on harmful reconstruction errors (RQ1)
Semestr 4	Further experiments on RQ1. Preparing and submitting the first publication to a 200-point venue in machine learning or robot learning, such as ICLR, RSS or IEEE RA-L. First experiments on RQ2
Semestr 5	Experiments on localizing the gap inside models and aligning representations (RQ2). Preparing a scientific publication
Semestr 6	Experiments on selecting real data (RQ3) and on transfer to unseen scenes and between tasks (RQ4). Preparing a publication and, if access allows, tests on a real robot
Semestr 7	Final experiments and submitting a paper to a conference or journal. First draft of the dissertation
Semestr 8	Completing and revising the doctoral dissertation, then submitting it

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission



Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

September 2029

(rekomenrowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Deep learning has made great progress in computer vision and language processing, largely because vast amounts of data could be collected from the web. Physical AI, meaning AI in robots and other systems that perceive and act in the real world, cannot rely on such data. Collecting interaction data with real robots is slow and expensive, and sometimes unsafe, so robot models are commonly trained in simulation. Recently, digital twins, which are simulations built from real data, have become a promising alternative to hand-crafted simulators. Neural scene reconstruction methods such as 3D Gaussian Splatting turn a short recording of a real place into a photorealistic copy. A model can be trained in this copy and then transferred back to reality.

However, a model that performs well in its digital twin often fails in reality, and even more often in places it has never seen. No twin reproduces the world exactly. Appearance, geometry, lighting and physics are only approximated, and reconstruction errors differ between regions of a scene. The resulting drop in performance is called the simulation-to-reality gap. It is a case of learning under distribution shift and one of the main obstacles to physical AI.

Current methods address it in part. Domain randomization varies a few global simulator parameters, regardless of where the twin is wrong. Invariant representations may discard information that the task needs. The few real data collected for adaptation usually correct either the model or the simulation, but not both. Most methods are also evaluated on a single task and a few scenes, so it is still unclear which properties of a twin and of the learned representations decide whether a model will work in reality.

My research aims to reduce this gap with methods that analyse the representations learned by the models and align them at the stage where the gap arises. I want to find out which reconstruction errors of a twin harm a model and at which stage inside the model task information is lost on real inputs. I also want to find the small set of real data that best corrects both the twin and the model, and to check whether the improvements carry over to unseen scenes and from one task to another. I will study two tasks of equal weight, robot navigation and robotic manipulation, and measure all results on scenes that were not used for training.

The results could be applied in service, logistics and assistive robotics. A robot could be trained in a digital copy of a new warehouse, hospital or home and then work there reliably after little real-world adaptation. They also apply to autonomous driving and inspection robots, whose models are likewise trained on simulated or reconstructed data.



6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / Outline of the current state of research in the topic of the doctoral dissertation

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Deep learning models for robot navigation and manipulation need more experience than real robots can provide, so they are usually trained in simulation. A study on navigation showed that performance in simulation can poorly predict performance in reality unless the simulator is carefully tuned [1]. The simulation-to-reality gap is an instance of distribution shift. Domain adaptation theory bounds the error of a model on the target domain by its error on the source domain, the discrepancy between the two distributions and the joint error of the best single model on both domains [2]. I treat this bound only as motivation. It suggests making the training distribution cover reality and making the model insensitive to the remaining discrepancy. Invariance does not target the joint error, however, and can even increase it, while correcting the simulation towards reality can reduce this error directly.

A widely used approach is domain randomization, which varies simulator parameters such as textures so that reality appears to the model as one more variation [3]. Randomizing too widely leads to conservative behaviour, so later works shape the distribution by fitting it to a few real rollouts [4] or by maximizing its entropy while preserving task success [5]. These methods tune a few global parameters of a synthetic simulator. They do not handle errors that vary across a scene, which are typical of twins reconstructed from real data.

Another approach makes the model invariant to the domain. Domain-adversarial training makes the features of the two domains indistinguishable to a domain classifier [6]. In robotics, aligning the joint distributions of observations and actions in simulated and real data improved co-trained policies in reality [7]. A policy here is a model that maps observations to actions. However, invariance with low source error does not guarantee transfer. It can increase the joint error term when the label distributions, here the actions, differ between domains [8], and full invariance may discard task information. A recent analysis showed that effective co-training on simulated and real data aligns the two domains while keeping them distinguishable [9], so separability of the domains alone does not show where the gap harms the task. Invariance is also usually enforced at locations chosen in advance, such as the input, the final features or fixed layers, without first measuring where the gap arises.

Digital twins built from real data are a more recent direction. 3D Gaussian Splatting [10] reconstructs photorealistic scenes from multi-view photographs and renders them in real time. In manipulation, twins have been used to train policies from a scan of the target scene [11], to transfer image-based policies to reality without real-world demonstrations [12] and to guide real-world reinforcement learning from a phone capture [13]. The SIMPLER study [14] showed with paired simulated and real evaluations on two robot setups that scenes matched to real images, with matched controllers, track the real-world performance of the same policies. In navigation, policies have been fine-tuned in twins built from a phone capture of a room [15] and trained in twins built from monocular urban videos [16]. Simulation factors have mostly been studied globally, for example randomization, photorealism and physics realism in real manipulation [17]. For whole workspaces, the similarity of pretrained features between rendered and real views predicted the agreement of simulated and real performance better than LPIPS or PSNR [18]. It is still open which local reconstruction errors harm generalization in navigation and manipulation, and how training variation should follow the reconstruction error and task relevance of each region.



It is also unclear where inside a model the gap arises. Probing, which reads out information from hidden states with simple classifiers, showed that fine-tuning for actions degrades the visual representations of vision-language-action models [19]. Separately, fine-tuning only selected layers showed that the best layers to adapt depend on the type of shift [20]. To my knowledge, in models trained in twins, nobody has located the stage at which task information available in simulation stops being recoverable from real inputs.

The remaining gap is usually reduced with a small amount of real data. Co-training with a small real set gives large gains [21]. Active system identification designs an exploration policy in simulation so that its real trajectory tells as much as possible about the physical parameters [22]. The work closest to mine uses a twin to find failure-prone configurations for real rollouts [13]. These works use the selected real data to correct either the model or the simulation. To my knowledge, correcting both from the same selected data has not been studied.

Finally, the gap can be split into factors of variation whose difficulty ordering is largely the same in simulation and reality [23]. Few studies test whether measurements taken before a method is applied can predict whether its benefit carries over to a new scene or to another task better than simple indicators such as the size of the gap. My dissertation addresses these questions.

(Please see section 12 for citations used in this document)

7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

The aim of my dissertation is to develop representation learning methods that improve the generalization of deep learning models trained in digital twins built from real data. The models should work reliably in reality and in places not seen during training, and the methods should not depend on a particular robot, scene or task.

I will not have my own robot at the start and will train models on academic GPU clusters, so the methods must work at this scale. I will rely on public datasets that contain independent captures of the same real scenes. I will study two tasks, robot navigation and robotic manipulation, and measure all results on held-out scenes that were not used for training or model selection.

My research questions (RQ) and hypotheses (H) are as follows.

1. RQ1. Which reconstruction errors of a digital twin, varying across a scene, harm the generalization of navigation and manipulation models trained in it?
2. H1. The harm caused by an error in a region is predicted better by how much it changes the features of a fixed pretrained visual encoder on views rendered with and without the error than by its size in the image or geometry, as measured by PSNR, LPIPS or depth error. Also, randomizing each region according to its estimated reconstruction error and task relevance generalizes better than uniform randomization of equal total strength.
3. RQ2. At which stage of a model trained or fine-tuned in a twin do real inputs first lose task information that is available for simulated inputs? I expect linear probes checked against control tasks to identify an earliest stage or a narrow range of stages. I will test with causal interventions whether the model uses this information.
4. H2. Aligning simulated and real representations at that stage reduces the simulation-to-reality gap more than the same alignment at the input, at the final features, at all stages or at layers chosen by existing criteria, given the same real data and training effort.
5. RQ3. Which real data, chosen under a fixed budget with the representation-based signals of RQ1 and RQ2, best correct both the digital twin and the model? I expect that correcting both with the



same selected data reduces the gap more than correcting only one of them. I will compare my selection with random and failure-driven selection.

6. RQ4. Do the gains transfer with unchanged settings to unseen scenes and from one task to the other? Can properties of the shift measured beforehand predict this better than simple indicators such as the size of the gap or the image-level discrepancy?
7. Fallback. If harm tracks the size of the error more closely than the change in features, I will guide randomization and data selection with reconstruction uncertainty and action sensitivity instead. Uncertainty misses errors that the reconstruction reproduces consistently, so I will use this fallback only if the representation-based signals turn out to be uninformative.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The main expected contribution of my dissertation is a set of representation learning methods that help deep learning models trained in digital twins generalize better to reality and to unseen places. I also want to learn whether the simulation-to-reality gap comes mainly from a limited set of task-relevant discrepancies that can be found among the reconstruction errors of a twin, located inside the model and reduced with a small amount of well-chosen real data.

The experiments should indicate which reconstruction errors harm generalization, at which stage of a model task information is lost on real inputs and when it pays off to correct both the twin and the model with the same real data. They should also show whether the gains carry over to new scenes and between navigation and manipulation. These questions concern learning under distribution shift, so the findings may also matter for domain adaptation, representation learning and active learning in general. I plan to release the code, the evaluation protocol and, where licences allow, the trained models, so that others can compare methods on held-out scenes in the same way. I hope the dissertation will bring machine learning research on distribution shift closer to robotics practice and make it easier to deploy learning-based robots in new warehouses, hospitals and homes.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Most of my work will consist of building digital twins of real scenes, training and fine-tuning deep learning models in them, and designing methods that analyse and align the representations these models learn. No established theory guarantees generalization from simulation to reality, so the work will be mostly empirical. Domain adaptation theory will be my motivation, and where possible I will analyse the properties of my methods theoretically.

I will use PyTorch for model training, gsplat for 3D Gaussian Splatting [10], COLMAP for camera poses, the Habitat simulator for navigation and ManiSkill3 for manipulation. I will use Git with GitHub repositories for version control and Hugging Face for downloading and publishing models and datasets. I will start from pretrained models where possible. I will release the code and evaluation scripts, and also trained models and derived data where licences allow.

Since I will not have my own robot at the start, I will evaluate mainly on public data. For navigation I will use public datasets of real indoor scenes with independent captures of the same scenes, such as



ScanNet++ and MuSHRoom. I will build the twin from one capture and use the other for real frames and a dataset-based reference. This reference is a proxy for reality. It measures the gap caused by reconstruction fidelity and leaves out differences in actuation and sensors. For manipulation I will build twins of real tabletop scenes from images in public robot datasets, following the visual matching of SIMPLER [14], and compare them with published real-robot results of the same policies. If I get access to a robotics laboratory at the university, I will use a real robot to check the direction of selected results and the validity of the proxy.

For RQ1 I will correct one type of reconstruction error at a time in selected regions using a more accurate reference, such as the laser scans of ScanNet++, or inject it in a controlled way, and measure the effect on the trained model. I will compare the change in the features of a fixed pretrained encoder with PSNR, LPIPS and depth error as predictors of harm and test region-wise randomization against uniform randomization of equal total strength (H1). For RQ2 I will train linear probes with control tasks on simulated hidden states and test them on paired real ones. Causal interventions on hidden states, such as patching the probed subspace from simulated into real inputs, will show whether the model uses the lost information. I will then add an alignment loss at the identified stage and compare it with alignment at the input, at the final features, at all stages and at layers chosen by existing criteria (H2). For RQ3 I will compare rules for selecting real data with random and failure-driven selection at equal budgets, and compare correcting both the twin and the model with correcting only one. For RQ4 I will apply the methods with unchanged settings to held-out scenes and check whether measurements taken beforehand predict the gains, using rank correlation with appropriate significance tests.

On held-out scenes I will measure the success rate and the simulation-to-reality gap, which is the difference between the performance of the same model in the twin and in reality or its reference. I will compare methods with relevant baselines under comparable training effort and real data. To get reliable results I will repeat runs over random seeds and scenes, use statistical tests for comparisons and run ablation studies. Data for training, model selection and final evaluation will be kept separate.

I will run experiments on the supercomputers of the Wrocław Centre for Networking and Supercomputing (WCSS), with jobs managed by SLURM, and on PLGrid resources. I intend to use AI coding assistants like Claude Code to speed up writing and testing code. They will also help with routine work such as monitoring long experiments and proofreading.

I plan to publish my results at conferences and in journals worth 200 points on the ministerial list. In machine learning these are NeurIPS, ICML and ICLR, in computer vision CVPR, ICCV and ECCV, and in robot learning RSS and the journal IEEE RA-L. Where possible, I will also post the papers as arXiv preprints.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Roboty coraz częściej uczą się w cyfrowych bliźniakach, czyli wirtualnych kopiach rzeczywistych miejsc odtworzonych na podstawie ich nagrań, ponieważ nauka w prawdziwym świecie jest powolna, kosztowna i czasem niebezpieczna. Taka kopia nigdy nie jest idealna. Na przykład szklane drzwi mogą zostać odtworzone błędnie i zmylić robota. W efekcie model, który dobrze działa w symulacji, w rzeczywistości często zawodzi, zwłaszcza w miejscach, których wcześniej nie widział. Chcę ustalić,



które różnice między kopią a rzeczywistością szkodzą modelowi i w którym miejscu wewnątrz modelu zamieniają się w błędy. Chcę też sprawdzić, które nieliczne rzeczywiste przykłady najlepiej poprawiają zarówno model, jak i kopię. Zbadam to w nawigacji robotów oraz w chwytaniu i przedstawianiu przedmiotów. Efektem mają być metody, dzięki którym roboty będą mogły szybciej, taniej i bezpieczniej uczyć się pracy w nowych miejscach.

Abstract for general public

Robots are increasingly trained in digital twins, that is, virtual copies of real places rebuilt from their recordings, because learning in the real world is slow, costly and sometimes unsafe. Such a copy is never perfect. For example, a glass door may be rebuilt incorrectly and mislead the robot. As a result, a model that works well in simulation often fails in reality, especially in places it has not seen before. I want to find out which differences between the copy and reality harm the model and where inside the model they turn into errors. I also want to check which few real examples best correct both the model and the copy. I will study this in robot navigation and in grasping and moving objects. The result should be methods that let robots learn to work in new places faster, more cheaply and more safely.

11. Termin oddania do druku artykułu naukowego lub monografii / [Date of submission for publication of at least one scientific article or monography](#)

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / [A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act \(Art. 267 sec. 2 point 2\(b\) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act \(Art. 267 sec. 2 point 2\(a\), referred to in Art. 186 sec. 1 point 3 of the Act.](#)

September 2027 (planned submission to a venue worth 200 points on the ministerial list, e.g. the ICLR 2028 conference or the journal IEEE Robotics and Automation Letters).

12. Inne / [Other comments](#)

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / [Additional explanations, comments, summary related to the individual research plan \(at most 1 page, font 11, line spacing 1\).](#)

I prepared this plan with the help of Claude, an AI tool that I used through Claude Code. I used it to search and check the literature, draft text, proofread, edit and format. I reviewed all content myself, and the references were checked against publisher and preprint records.

Bibliography

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13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

My supervisor and I plan weekly meetings, where I will present my progress and recent experimental results, we will agree on the next steps and we will work on joint papers. I will also attend the weekly meetings of my supervisor's group of PhD students, where we discuss current results, problems and new work in the field. For daily contact we will use email and instant messaging. Code, trained models and the LaTeX sources of papers will be kept in shared Git repositories.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor



Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor